PART 1: USER PROFILE & ACCOUNT INITIALIZATION

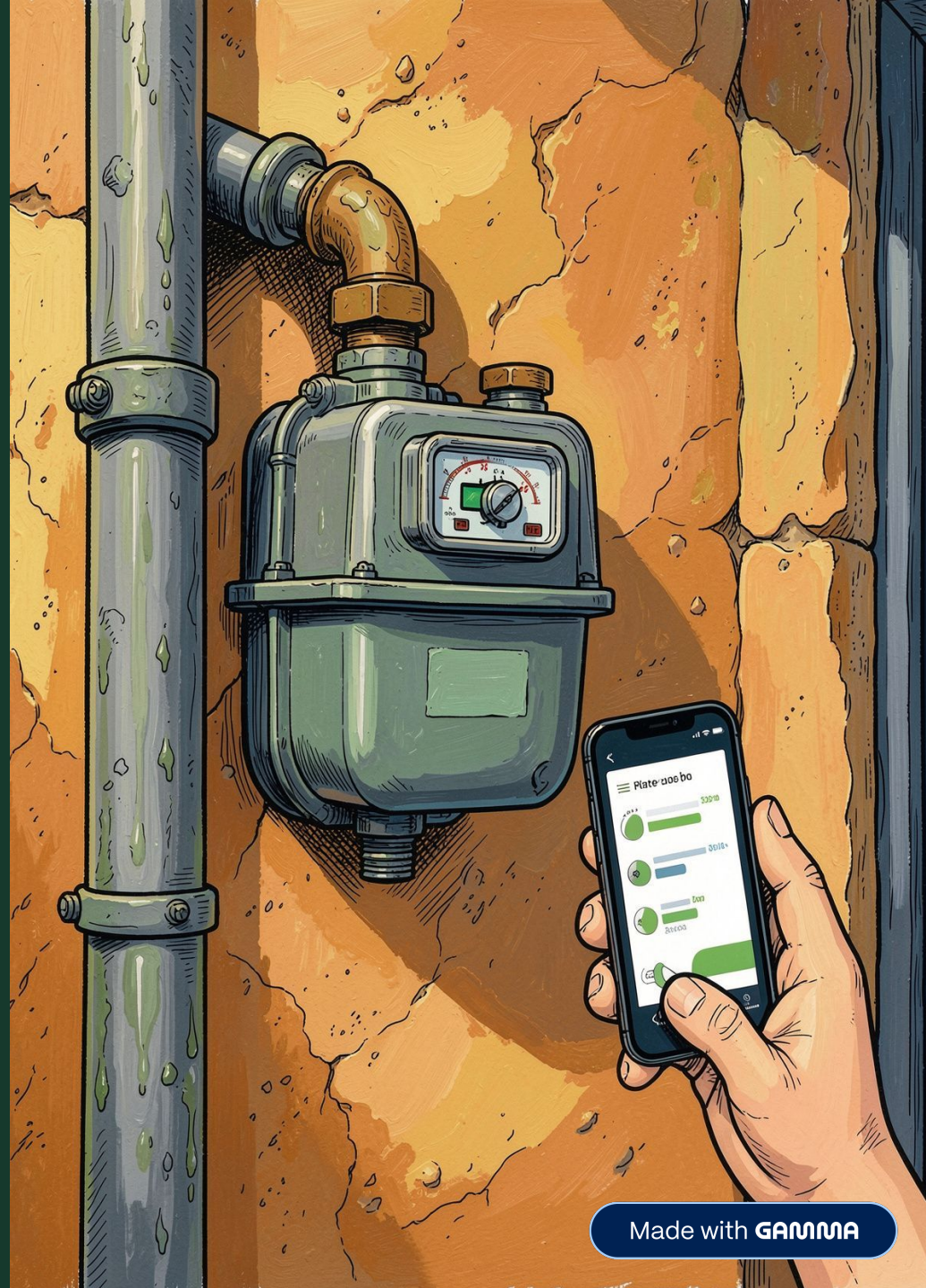
Household Water-Usage & Billing Monitor

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Course: Java Programming

Module 1



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PART 1

User Profile & Account Initialization

PROJECT INTRODUCTION

Project Introduction



The project is designed to monitor household water usage.



It collects user details and water meter readings.



It calculates water consumption and generates bills.



The system stores information securely for future use.



This part focuses on creating and initializing the user account.

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OBJECTIVE

Objective of Part 1

User Profile & Account Initialization

1. Collect customer information.
2. Create a user account.
3. Store account details using a Java class.
4. Protect data using encapsulation.
5. Initialize the account with valid information.

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USER INFORMATION

User Information

- Customer Name
- Roll Number / Customer ID
- Contact Number
- Account details

Example

FIELD	EXAMPLE
Name	Rahul
Roll No.	24A91
Contact	9876543210

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OOP CLASS DESIGN

OOP Class Design

We create a Java class called UserProfile.

```
class UserProfile {
    private String name;
    private String rollNo;
    private String contact;

    public UserProfile(String name, String rollNo, String contact) {
        this.name = name;
        this.rollNo = rollNo;
        this.contact = contact;
    }
}
```

Key concept: The class represents a customer's profile, while the object stores the actual customer's information.

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OOP CONCEPT

Encapsulation

Encapsulation means wrapping data and methods together inside a class and controlling access to the data.

In our project:

- User information is declared as private.
- Direct access from outside the class is restricted.
- Getters and setters can be used to access or modify the data.

Easy way to remember: 🔒
Private data + controlled access = Encapsulation

```
private String contact;

public String getContact() {
    return contact;
}

public void setContact(String contact) {
    this.contact = contact;
}
```


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INITIALIZATION

Account Initialization

When a new customer joins the system:

Input → Object Creation → Account Initialized

```
UserProfile user1 =
    new UserProfile("Rahul", "24A91", "9876543210");
```

UserProfile → class

user1 → object

"Rahul" → customer name

"24A91" → roll number

"9876543210" → contact number

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PROCESS

Working Flow

01	02	03	04	05
Customer Details	Create UserProfile Object	Initialize Account	Store Data Securely	Ready for Meter Reading

This connects Part 1 to Part 2, where the user's water meter readings will be collected.

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BENEFITS

Advantages



Keeps customer information organized.



Protects important data.



Makes the program easier to maintain.



Demonstrates OOP concepts.



Allows each customer to have a separate profile.



Provides a foundation for the remaining modules.

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CONCLUSION

Conclusion

- ✓ We created a basic user profile for the water billing system.
- ✓ Customer information is stored using a Java class.
- ✓ Encapsulation helps protect the data.
- ✓ Objects are used to initialize individual accounts.
- ✓ The account is now ready for the next stage: Meter Reading & Ingestion.

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